

Properties in SR

Item properties

Item IDs

Side information

Multi-modal features (e.g., text descriptions, images, videos, ...)

General features

Categorical features (e.g., brands, categories, colors, domains, ...)

Numerical features (e.g., prices, ...)

Knowledge graph

User properties

User IDs

Side information

Multi-modal features (e.g., text descriptions, images, ...)

General features

Categorical features (e.g., genders, ...)

Numerical features (e.g., ages, heights, ...)

Social networks

Behavior properties

Multi-modal features (e.g., reviews, ...)

General features

Behavior types

Ratings

Context information (e.g., timestamps, locations, weather, ...)

...

Sequence properties (e.g., lengths, internal order, ...)